Asymptotic Notations:

* It describes the behaviour of the time or space complexity for large instance characteristics of a program
* While analysing an algorithm, 3 notations are used namely
  - i) Big Oh Notation - O
  - ii) Big Omega Notation - Ω
  - iii) Theta Notation - Θ

i) Big Oh Notation - O

Let f and g be 2 functions defined from a set of natural numbers to the set of non-negative real numbers.

(ie) $f, g : N \rightarrow R_{\geq 0}$ $\qquad C \in R,\ n_0 \in N$

$f(n) = O(g(n))$, iff there exists positive constants c and n_0, such that $f(n) \leq c * g(n)$ for all n, $n \geq n_0$

* This defines the upperbound for a function.

Ex: $f(n) = 8n + 128$

$g(n) = n^2$

$f(n)$ is non-negative for all integers, $n \geq 0$

To prove: $8n + 128 = O(n^2)$

To find an integer n_0 and $c > 0$ such that for all integer $n \geq n_0$, $f(n) \leq cn^2$

Assume $c = 1$

then $f(n) \leq n^2$

$$8n + 128 \leq n^2$$

$$0 \leq n^2 - 8n - 128$$

$$0 \leq (n+8)(n-16)$$

$$(n+8) > 0 \quad \forall \; n \geq 0$$

$$(n_0 - 16) \geq 0$$

$$n_0 = 16$$

$\therefore$ for $c = 1$ & $n_0 = 16$, $f(n) \leq c \cdot n^2 \; \forall \; n \geq n_0$

ii) Big Omega Notation - Ω

Let f and g be two functions defined from a set of natural numbers to a set of non-negative real numbers.

(ie) $f, g : N \rightarrow R_{\geq 0}$ $\qquad c \in R, \; n_0 \in N$

$f(n) = \Omega(g(n))$, iff there exists positive constants c and n_0, such that $f(n) \geq c * g(n)$ for all n, $n \geq n_0$.

* This defines the lowerbound for a function.

Ex

$f(n) = 5n^2 - 64n + 256$

$g(n) = n^2$

To prove: $f(n) = \Omega(g(n))$ $\therefore$ $f(n) \geq c(g(n))$

Proof:

$f(n)$ is always non-negative for all integers, $n \geq 0$

$(5n^2 - 64n + 256) \geq c * n^2$

Assume $c = 1$,

$5n^2 - 64n + 256 \geq n^2$

$4n^2 - 64n + 256 \geq 0$

$n^2 - 16n + 64 \geq 0$

$(n-8)(n-8) \geq 0$

$(n-8)^2 \geq 0$

$\because (n-8)^2 > 0$ for all values $n > 0$,

$\Rightarrow n_0 = 8$

$\therefore$ for $c = 1$ & $n_0 = 8$, $f(n) \geq c \cdot n^2$ for all integers; $n \geq n_0$

iii) Theta Notation - Θ

Let f and g be two functions defined from a set of natural numbers to a set of non-negative real numbers.

(ie) $f, g : N \rightarrow R_{\geq 0}$ $\qquad$ $c_1, c_2 \in R$ & $n_0 \in N$

$f(n) = \Theta(g(n))$, iff there exists 3 positive Constants $c_1, c_2 \in R$ & $n_0 \in N$ such that

$$c_1 g(n) \leq f(n) \leq c_2 g(n) \quad \forall \quad n \geq n_0.$$

*This defines both the upper & lower bound for a function.

* $f(n)$ is theta $g(n)$ iff $f(n)$ is $O(g(n))$ & $\Omega(g(n))$

$3n+2 = \Theta(n)$ $\qquad$ $f(n) = 3n+2$ $\quad$ $g(n) = n$

$3n+2 = 3 \cdot n \quad \forall \; n \geq 2$

$3n+2 = 4n \quad \forall \; n \geq 2$

$c_1 = 3 \quad c_2 = 4$ & $n_0 = 2$.